

Heart Disease Prediction with Classical ML Models

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Abstract—This project investigates the design of an end-to-end machine learning pipeline for binary classification on a large and heterogeneous health-related dataset. The objective is to predict coronary heart disease (MICHHD) from BRFSS survey data. We implement six baseline algorithms from scratch, design a compact preprocessing stack, and select models via 5-fold cross-validation.

I. INTRODUCTION

Cardiovascular diseases are a major public health issue. Given tabular health and lifestyle features from BRFSS, we address a binary classification task ($y \in \{0, 1\}$) to estimate MICHHD risk. Per course constraints, we use only Python standard libraries and NumPy (no external ML libs). Our contributions are: (i) re-implementations of six classical methods; (ii) a concise, leakage-safe preprocessing; (iii) an ablation-driven selection showing why regularized logistic regression is a robust default on this dataset.

II. IMPLEMENTATION OF THE 6 MODELS

- **Mean Squared Error (GD).** This method performs linear regression using gradient descent, iteratively adjusting the model weights to minimize prediction error. It is simple, effective, and suitable for moderate-sized datasets.
- **Mean Squared Error (SGD).** The stochastic variant updates weights using one random sample at a time. It is faster and more scalable on large datasets but introduces some variability in convergence.
- **Least Squares.** This analytical approach directly computes the optimal weights without iteration. It serves as a reliable baseline for linear regression and allows performance comparison with gradient-based methods.
- **Ridge Regression.** Ridge regression adds a regularization term that limits the impact of large coefficients. This helps reduce overfitting and improves model stability, especially when features are correlated.
- **Logistic Regression.** Used for binary classification, this model estimates the probability of belonging to one class rather than another. It is trained by minimizing a loss function derived from likelihood estimation.
- **Regularized Logistic Regression.** This variant includes an additional penalty to prevent overfitting and improve generalization. It is particularly effective on noisy, high-dimensional, or unbalanced datasets.

III. DATA & PREPROCESSING

Our preprocessing pipeline was designed to remove unreliable information, standardize all remaining variables, and

prevent any data leakage from training into validation. Each transformation was fitted exclusively on the training fold and subsequently applied to the validation and test data.

Before finalizing the thresholds used at each stage (NaN ratio, constant ratio, correlation threshold, oversampling ratio), we conducted multiple experiments by training our best-performing model across several configurations. The final cutoffs were selected based on the values that yielded the most stable and accurate validation performance.

- **Manual exploration and variable pruning.** We began by manually inspecting the dataset to identify non-relevant or redundant variables. Administrative and survey-related fields (e.g., FMONTH, IDATE, SEQNO) were removed, as they contained metadata or logistical information rather than genuine health signals. This first cleaning ensured that only variables with potential predictive value were preserved and avoided biases linked to data collection artifacts.
- **Handling missing values.** Columns with excessive missing data were dropped to prevent unreliable imputations. Specifically, we retained only features with less than 80% missing values, as columns with higher ratios rarely contribute useful information. Table I summarizes the number and percentage of dropped columns across several thresholds, showing that a cutoff around 80% provides a strong compromise between dataset completeness and feature reliability.

TABLE I
EFFECT OF MISSING-VALUE THRESHOLDS ON FEATURE RETENTION.

NaN thr	Drop	Keep	%Drop
0.50	147	165	47.12
0.60	129	183	41.35
0.70	121	191	38.78
0.80	116	196	37.18
0.90	99	213	31.73

- **Removing constant features.** We also discarded features with negligible variance, i.e., those where more than 99% of values were identical. Such variables contribute no discriminative information and can distort normalization. Table II presents the effect of different thresholds. A cutoff at 99% removed only a few redundant variables while preserving the majority of informative features.
- **Correlation filtering.** To prevent multicollinearity, we computed pairwise Pearson correlations and removed one variable in each pair with $|\rho| \geq 0.95$. This greedy filtering

TABLE II
EFFECT OF CONSTANT-RATIO THRESHOLDS ON FEATURE PRUNING.

Const thr	Drop	Keep	%Drop
0.800	55	141	28.06
0.850	43	153	21.94
0.900	26	170	13.27
0.950	13	183	6.63
0.990	7	189	3.57

keeps only one representative feature per correlated cluster, stabilizing model training and improving interpretability after encoding.

- **Feature typing and imputation.** Features were categorized as continuous or categorical based on their number of unique non-missing values. After manual inspection, we determined that features with at most 15 distinct values should be treated as categorical. This range captured most discrete attributes while continuous features naturally exhibited a higher cardinality. Missing values were imputed from the training fold only: continuous variables with their median (robust to outliers), and categorical ones with their mode (preserving the most frequent category).
- **Scaling and encoding.** Continuous features were standardized to zero mean and unit variance, ensuring comparability across variables and improving gradient-based optimization. Categorical features were one-hot encoded using specifications fitted on the training set, while the last level was dropped to avoid perfect multicollinearity after encoding.
- **Class imbalance correction.** Because only about 9% of samples belonged to the positive class, we applied a controlled oversampling of the minority class until it reached approximately 20% of the majority class size. The objective was to strengthen the model’s sensitivity to rare outcomes without overfitting duplicated instances.

IV. EXPERIMENTS AND RESULTS

- **Model and hyperparameter selection.** Models are evaluated using Accuracy and F1-score, with priority given to F1 to reduce false negatives, which are critical in medical prediction. Each model is trained with a fixed maximum of 1000 iterations to ensure consistent convergence, and hyperparameters are tuned via 5-fold cross-validation. Model selection is based on the average F1-score across folds, aligning the evaluation directly with the classification objective. Learning rate (γ) and regularization strength (λ) are optimized to maximize F1, followed by threshold adjustment on validation folds to handle class imbalance. The Regularized Logistic Regression model achieved the highest cross-validated F1 and was retained as the final model.
- **Validation metrics.** The validation results highlight the importance of using a classification-oriented metric such as F1. Regularized Logistic Regression achieves the highest F1, confirming that mild L_2 regularization

improves generalization and stability on noisy, imbalanced data. Linear regression-based methods (GD, LS, Ridge) perform comparably but remain less calibrated, as their optimization objective does not directly model class probabilities. These results indicate that optimizing the decision boundary through regularized logistic loss provides a more consistent and interpretable improvement for classification.

TABLE III
VALIDATION PERFORMANCE AND SELECTED HYPERPARAMETERS AFTER THRESHOLD TUNING

Model	γ	λ	Accuracy	F1
Gradient Descent	0.03	–	0.814	0.55
Stochastic GD	0.001	–	0.779	0.511
Least Squares	–	–	0.814	0.551
Ridge	–	10^{-3}	0.812	0.551
Logistic	0.5	–	0.814	0.55
Regularized Logistic	0.3	10^{-4}	0.823	0.555

Final model. We retain the regularized logistic regression with the validation-selected threshold and use it to produce the test submission. A validation confusion matrix for this setting (Fig. 1) indicates a balanced precision–recall trade-off consistent with the F1 optimum. This indicates that the classifier performs well overall, with strong detection of negative cases and a moderate ability to identify positives, though some positive instances are still missed.

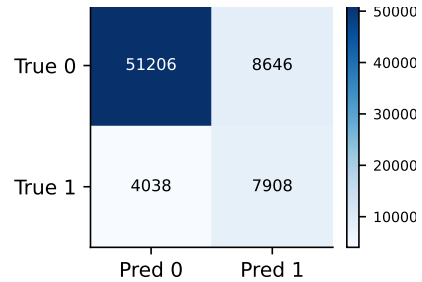


Fig. 1. Validation confusion matrix.

V. CONCLUSION

This project demonstrates that careful preprocessing and metric-aligned validation are key to achieving robust classification on noisy, imbalanced data.

The confusion matrix shows that while the model performs reasonably well overall, it still produces a substantial number of false negatives, an issue that is particularly critical in a medical context where missing positive cases can have serious consequences. Therefore, the model’s predictions should be interpreted with caution and not used in isolation for decision-making.

Future improvements could include more advanced resampling methods, automated feature selection or dimensionality reduction (e.g., PCA), and probability calibration to enhance decision reliability. Overall, this work highlights that strong performance stems less from algorithmic complexity than from disciplined data preparation and validation design.

REFERENCES

- [1] U. S. Department of Health and Human Services, “2015 BRFSS Survey Data and Documentation,” [Online]. Available: [Here](#)